

Hemanth Reddy

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EDUCATION

University of Maryland, College Park

Master of Science in Applied Machine Learning

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering in Mechanical Engineering CGPA: 8.31/10

College Park, Maryland, USA

Aug. 2026 – May. 2028 (Expected)

Hyderabad, India

Aug. 2020 – May. 2024

TECHNICAL SKILLS

Machine Learning & AI: LangChain, PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Sentence Transformers, Computer Vision, Deep Learning, NLP, LLMs, Generative AI, RAG, Information Retrieval, FAISS, LoRA, Stable Diffusion, GANs

Programming & Data: Python, C++, JavaScript, SQL, NumPy, Pandas, PostgreSQL, MATLAB

Cloud & ML Engineering: AWS, AWS Lambda, Amazon EC2, Amazon CloudWatch, Docker, FastAPI, Linux, Apache Kafka, Multiprocessing, Multithreading, Git

PROFESSIONAL EXPERIENCE

TELUS Digital

AI Software Development Engineer

Bengaluru, India

Jul. 2024 – Aug 2026

- Developed a **synthetic data generation pipeline** using **CycleGAN** and **Stable Diffusion** for **image-to-image translation**, generating photorealistic weather variants that supported **domain adaptation** and improved the robustness of downstream **autonomous vehicle perception models**.
- Engineered the migration of a legacy **computer vision pipeline** from **AWS EC2** to an event-driven **AWS Lambda** architecture, eliminating idle compute time and reducing infrastructure overhead by **\$60,000 annually**.
- Built a fault-tolerant **serverless framework** using **multiprocessing** and **thread pools** to accelerate CPU-heavy ML workloads, integrating **Amazon CloudWatch** and isolation boundaries to prevent pipeline failures.

Solutions Engineer Intern

Jul. 2023 – Jul. 2024

- Fine-tuned a **MarianMT Transformer** via **LoRA** on a **100K-sentence English-Nordic parallel corpus**, optimizing domain-specific machine translation for low-resource Nordic languages.
- Evaluated translation quality using the **COMET** metric, delivering a **15% accuracy gain** that measurably reduced downstream manual post-editing efforts.
- Developed a data-pruning pipeline using **DINOv2** embeddings and **cosine similarity** to strip out redundant images, cutting dataset volume by **70%** while preserving **90%** of the original semantic coverage.

Military College of Electronics and Mechanical Engineering

Project Intern

Secunderabad, India

May 2022 – Jul. 2022

- Fine-tuned a pretrained **YOLOv5** model to classify drones from birds using an annotated **1,500-image aerial dataset** and **image augmentation** techniques, achieving **87.2% mAP@0.5**.

PROJECTS

Self-Improving RAG Agent | *NLP, Generative AI, Information Retrieval* | github.com/RrHemanth/self-improving-rag

- Built an adaptive **RAG** agent combining **dense retrieval**, **BM25**, query rewriting, and **cross-encoder reranking** to improve context retrieval for LLM-based question answering.
- Implemented an **LLM-based evaluation and experience-memory loop** to diagnose retrieval failures and dynamically select improved retrieval strategies for future queries, enabling iterative agent improvement.
- Developed an evaluation benchmark measuring **retrieval accuracy**, **answer faithfulness**, and **relevance**, achieving a **25% improvement** over a baseline RAG system.

Video Anonymization Pipeline | *Computer Vision, Deep Learning* | github.com/RrHemanth/video-anonymization-system

- Built an end-to-end video anonymization pipeline using **YOLOv11** and **BoT-SORT** to detect, track, and anonymize faces and license plates across video sequences
- Improved temporal consistency through **track interpolation** and smoothing algorithms, reducing flickering caused by missed detections and unstable bounding boxes. Optimized and benchmarked the pipeline on an **NVIDIA Tesla T4 GPU**, reducing end-to-end inference time by **70%** over CPU execution.